

# Data Centers, Urban Heat, and AI Growth

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## Source Content

Data Centers, Urban Heat, and AI Growth As AI fuels data center expansion, communities are pushing back. What's driving the opposition, and what are the real environmental and economic trade-offs? YSE experts weigh in. As proposals to build data centers to meet the computing demands of artificial intelligence grow, so too does the pushback. Cities including Denver, Minneapolis, and Seattle are considering municipal restrictions on data centers, and according to Data Center Watch, a research project from technology company 10a Labs, more than 140 local groups have managed to block or delay more than \$60 billion in data center investments in roughly a year. The concerns around data centers are immediate and local: urban heat, water consumption, electricity grid strain, but the drivers are global and economic. Understanding the science, the political landscape, and the underlying economics is critical to understanding how this technology will shape communities. Here, Karen Seto, the Frederick C. Hixon Professor of Geography and Urbanization Science and Director of the Hixon Center for Urban Sustainability, explains what data centers actually do to urban heat. Daniel Esty, the Hillhouse Professor of Environmental Law and Policy, maps the cross-spectrum opposition and what it reveals about how society should approach AI's costs and benefits. Kenneth Gillingham, the Grinstein Class of 1954 Professor of Environmental and Energy Economics, breaks down the economic and energy realities driving the data center boom, and Yuan Yao, the Manufacturer's Association Professor of Industrial Ecology and Sustainable Systems, explains the environmental costs of data centers and what solutions exist to minimize their impact. Seto: Because nearly all of the electricity consumed by a data center ends up as heat, servers and other equipment require continuous cooling. These cooling systems remove the heat generated by computing and release it into the surrounding environment as waste heat. Recent research conducted in Phoenix, Arizona, estimates that a large data center can emit waste heat comparable to that produced by tens or even hundreds of thousands of households. The latest science shows that this waste heat can increase surrounding land surface temperatures by as much as 16°F and raise air temperatures in nearby neighborhoods by up to 4°F. These effects have been observed as far away as half a kilometer, or about five city blocks! While more research is needed, these findings suggest that data centers can worsen heat exposure in neighborhoods that are already impacted by the urban heat island effect. Given that cities are typically 1-7°F warmer than surrounding areas, additional warming from data center waste heat can further increase air conditioning energy demand and human health risks in nearby communities. In order to mitigate these impacts, cities and industry leaders should work together on better facility design and neighborhood cooling strategies, including parks and reflective surfaces. Esty: Reactions to the growth of data centers represent something of a Rorschach Test — perhaps telling us more about

the economic confidence of those offering opinions than anything else. Every week we see new AI-enabled tools emerging that might allow us to reconfigure our food, transport, building management, and energy systems for much greater efficiency and thus sustainability. AI also offers the potential to bring cutting-edge thinking and technologies to every corner of the globe at speed and scale, which might allow us to ramp up the global response to shared challenges such as climate change, biodiversity conservation, chemical exposures, and pollution. But AI also poses significant risks related to growing electricity demand that could cause a spike in greenhouse gas emissions as well as air and water pollution if the new power requirements are met by fossil fuels rather than clean power generation and improved energy efficiency. What has become evident is that society needs a more clear-eyed approach to the potential losses from AI as well as a strategy for sharing the benefits. Simply put, only with ongoing policy-guided investments and concerted effort can we be confident that AI will move society toward a sustainable future.

Gillingham: The main driver of data center expansion today is the race to train better AI models. In addition to training large language models, data centers are also used to run generative AI queries. They are also used simply to store all of the data that we all have in the cloud. But the primary driver of the growth really is generative AI. Generative AI may offer valuable benefits to society, but large data centers consume massive amounts of electricity and water. Some large data centers under construction are slated to use 3 GW or more of peak electricity. This is more than 75 times Yale's peak demand. We are beginning to see substantial anxiety about generative AI in the polls, as well as substantial pushback from some politicians. Regardless, energy policy will have to accommodate the first new growth in electricity demand in a couple of decades. Electricity prices are already rising, and they are likely to continue to increase in the near future. Higher electricity prices can help the adoption of renewables, but so far, the new electricity demand has largely been met with natural gas generation. This means increased emissions. We don't know how many new data centers will be built, but it is clear already that the next few years will be a very dynamic time for electricity markets and energy policy.

Yao: The environmental impacts of data centers are real, but they are also highly context-dependent. The most visible impact is electricity use, which can translate into greenhouse gas emissions depending on when and where the power is consumed. A data center located on a coal- or gas-heavy grid can have a very different carbon footprint from one using low-carbon electricity. Water use is another major concern, especially for cooling and for electricity generation upstream, and it matters greatly whether a facility is located in a water-stressed region. A life-cycle perspective is important because impacts do not come only from day-to-day operations. Data centers also require servers, chips, cooling equipment, buildings, backup power systems, and grid infrastructure, all of which carry embodied environmental impacts from manufacturing, construction, and replacement. The solutions that work best are not single fixes, and they may differ across locations, technologies, and operating models. Energy efficiency and improved cooling remain important, but efficiency gains can be offset if total demand grows rapidly. Clean electricity procurement is essential, but it should be evaluated by location and time, not just annual averages. More broadly, solutions need to consider siting and infrastructure decisions, operational strategies, and the full life cycle management of hardware and equipment. Transparent reporting and metrics that capture carbon, water, reliability, and life cycle environmental impacts together are also critical. My group is currently working on sustainability metrics for data centers, with the goal of helping companies, policymakers, and the public more clearly compare impacts across technologies, locations, and operational choices.

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